Deep VIO

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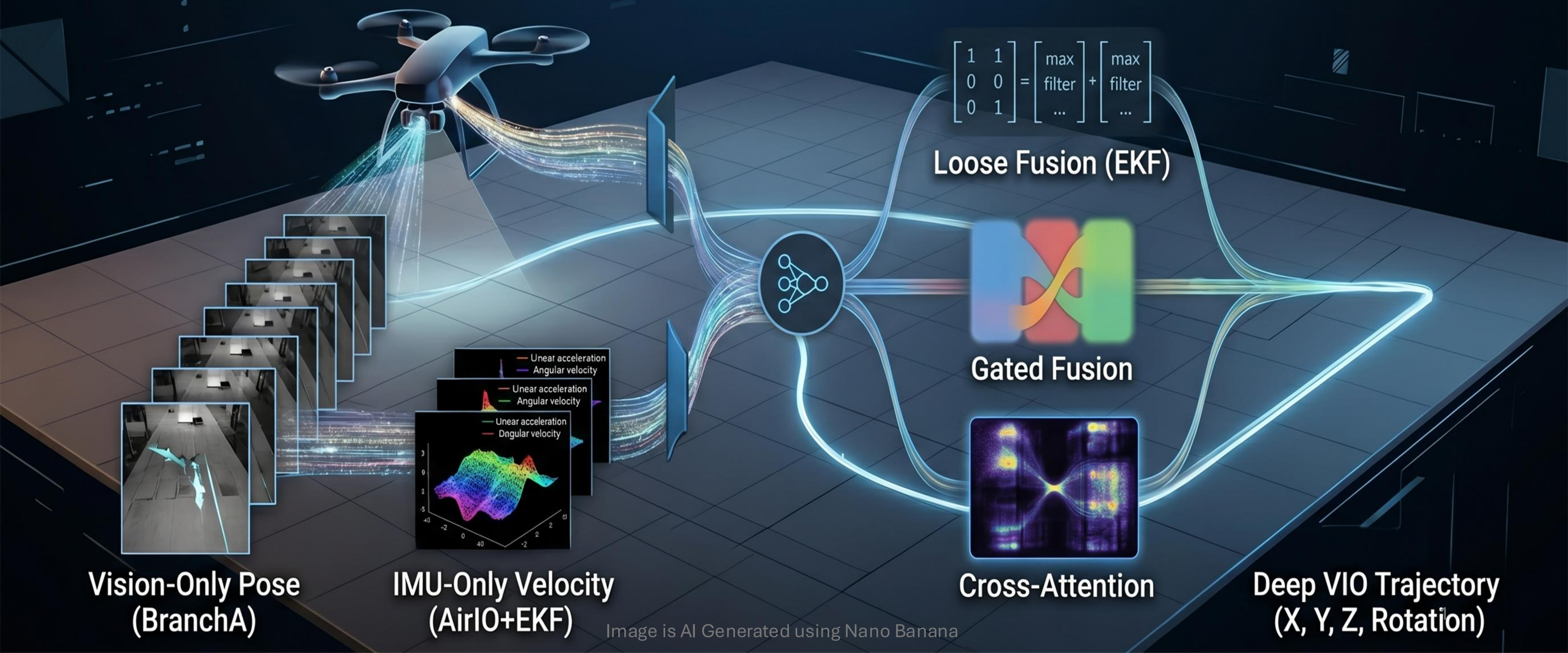


Image is AI Generated using Nano Banana

Data Generation

Trajectories:
Clover, Figure 8, Lemniscate, Oval
Images of various textures used:
City, Forest, Rocks, Grass, Marble
Blender used for rendering
IMU values generated using **GNSS-INS-SIM**

Initial tests resulted in higher errors in the z-axis. This was due to the most trajectories being in the x-y plane. Trajectories with verticality were introduced.

New Trajectories:
- Figure 8 (rotated in 3D)
- Clover with each leaf in 3D with sin function
- Lemniscate (rotated in 3D)
- Spiral (Axis along X-Y plane)
- Oval (left as is)

Vision Only Architecture

Stage 1

Siamese CNN Encoder

Stage 2

Correlation Layer

Stage 3

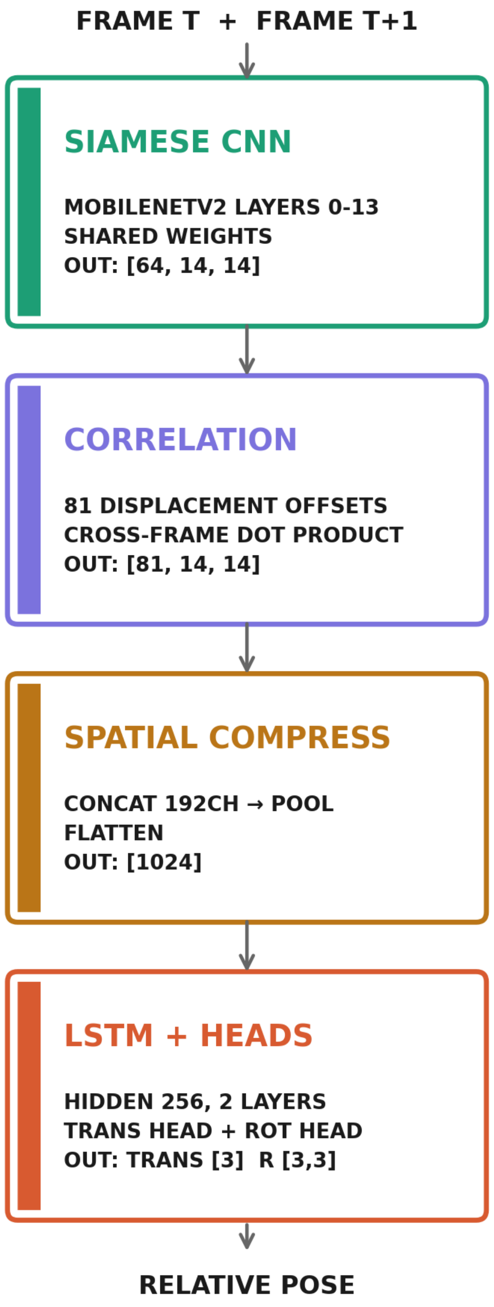
Spatial Compression

Stage 4

LSTM + Pose Heads

Loss

L1(trans) + 100 ×geodesic(rot)



DeepVO (Wang et al., ICRA 2017) · MobileNetV2 (Sandler et al., CVPR 2018) · PWC-Net (Sun et al., CVPR 2018) · 6D Rotation (Zhou et al., CVPR 2019)

Results and Observations

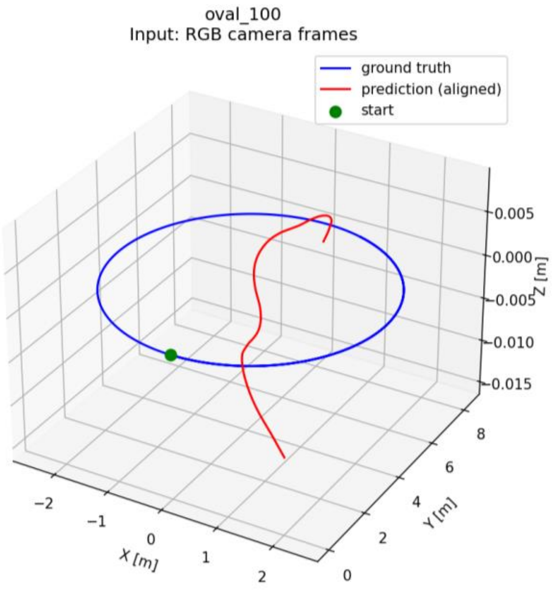
Camera only model had worse global drift than IMU
LSTM was expected to solve this drift issue, but the result was opposite.

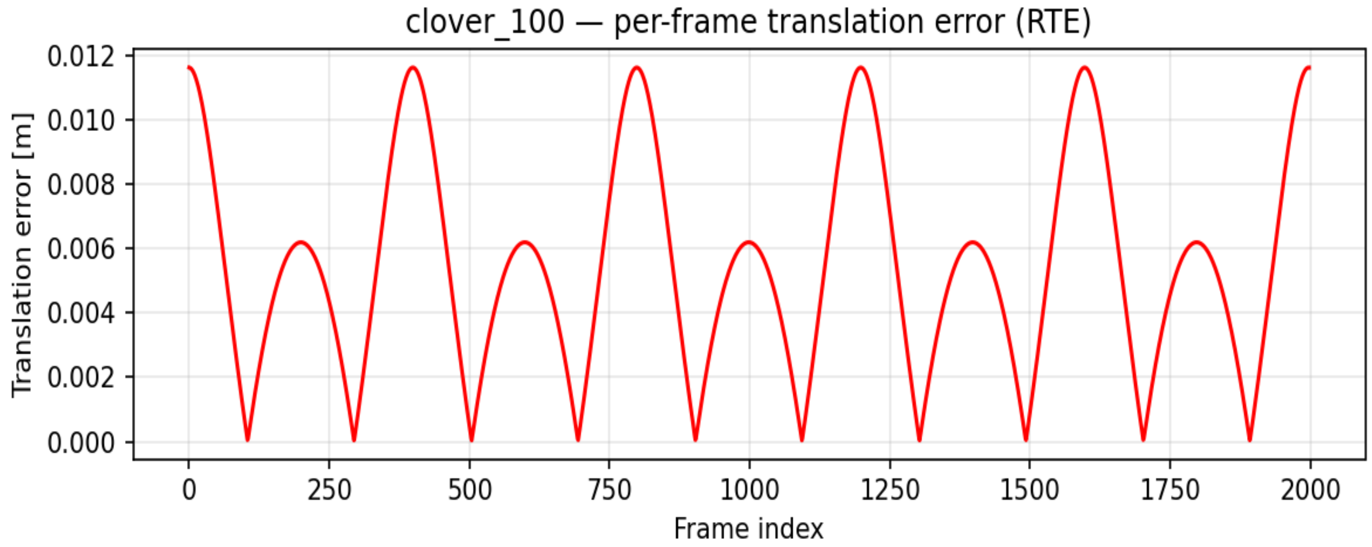
Per-frame RTE: 0.0056 m
Tiny step error. Looks like it is working.

Downward camera: yaw looks like translation
Rotating drone shifts ground texture identically to a sideways move.

LSTM does not fix this and remembers past frames but all past frames had the same wrong heading. This propagates through the training.

Global ATE: 2.85 m
Every wrong heading chains into the trajectory. The shape is completely lost.



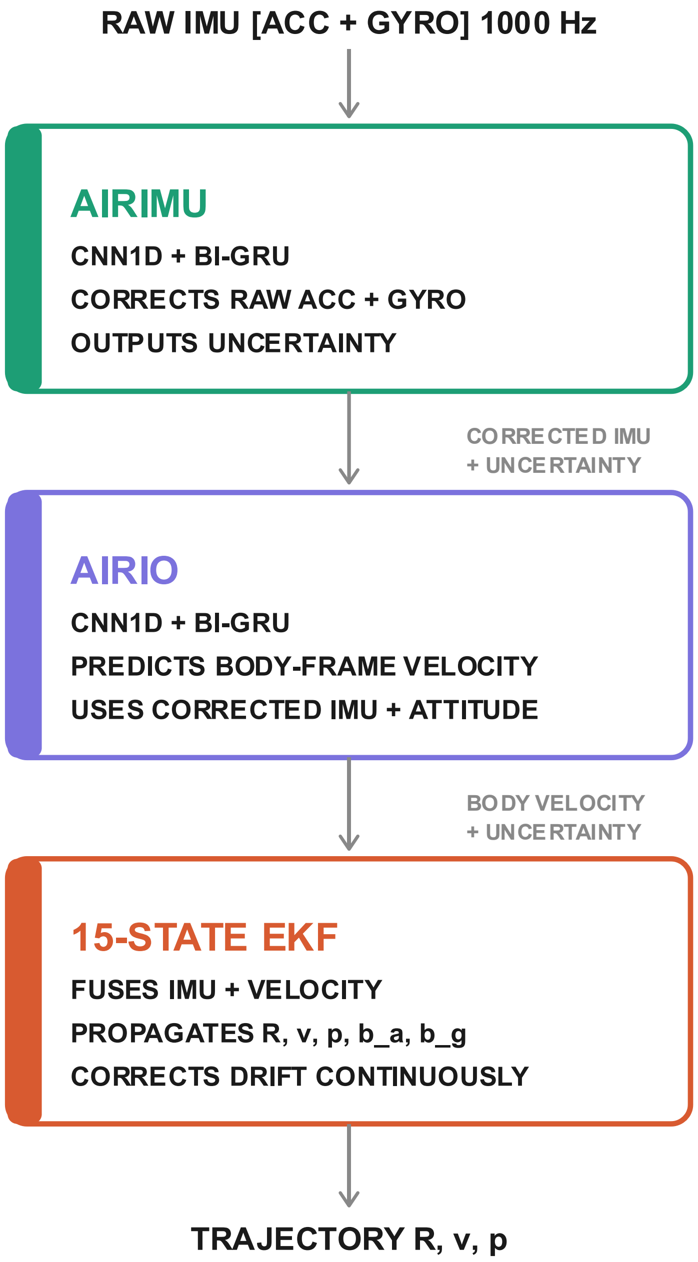


IMU gyroscope measures heading directly from physics. Vision must infer it from pixels.
For a downward camera, that inference fails at sharp turn.

DeepVO (Wang et al., ICRA 2017) · MobileNetV2 (Sandler et al., CVPR 2018) · PWC-Net (Sun et al., CVPR 2018) · 6D Rotation (Zhou et al., CVPR 2019)

IMU Only Architecture

Stage 1 — AirIMU Denoises raw IMU, outputs per-sample uncertainty
Stage 2 — AirIO Predicts body-frame velocity using corrected IMU and attitude
Stage 3 — EKF Fuses everything into a full 15-state trajectory estimate
Loss AirIMU: geodesic + MSE on pre-integrated window.
AirIO: Huber(velocity) + Gaussian NLL



AirIMU (Qiu et al., 2023) · AirIO (Qiu et al., RA-L 2025) · Forster et al., RSS 2015

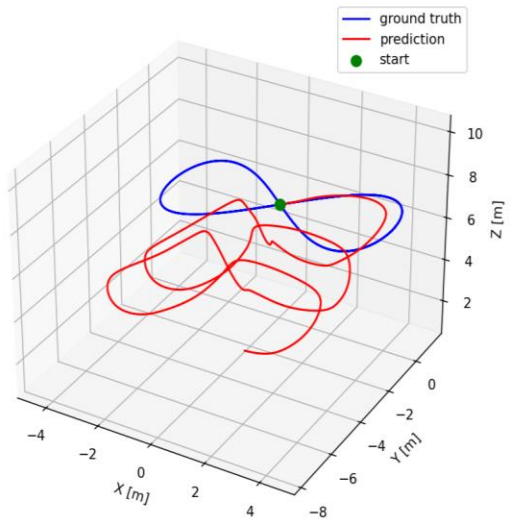
Results and Observations

It works when motion is familiar *Clover pattern seen in training → shape preserved, ATE 0.87 m*

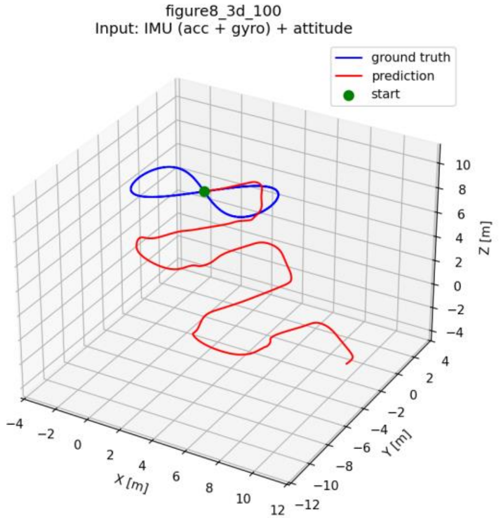
It breaks on unfamiliar motion *Figure-8 3D: novel axis combinations → prediction loses shape entirely*

Axis blindness + vz = 0 assumption *Network never sees vertical velocity → Z error accumulates every step*

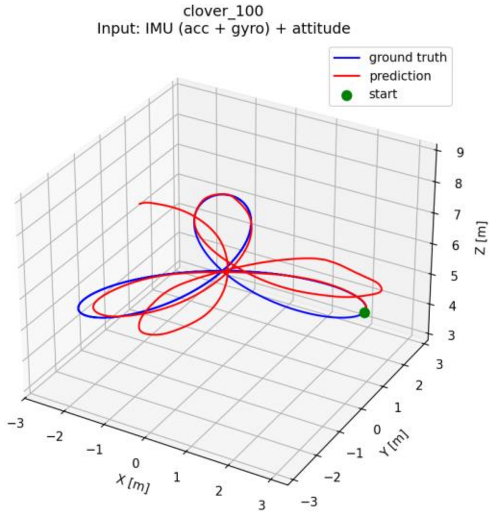
More data made it worse *Diverse sequences expose blind spots the model can't generalize across*



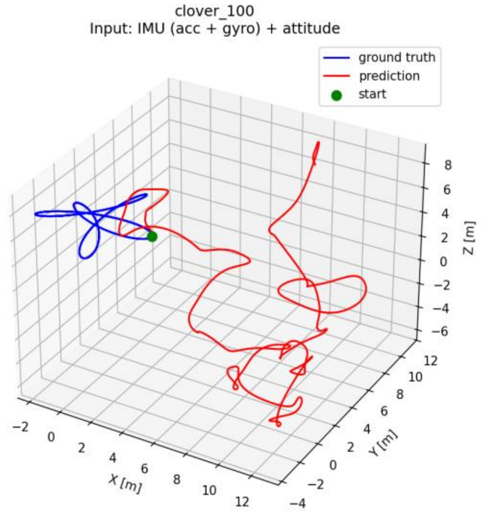
ATE: 4.23 m
RTE: 3.00 m
rot err: 23.6°



ATE: 8.55 m
RTE: 5.97 m
rot err: 60.1°



ATE: 0.78 m
RTE: 0.52 m
rot err: 14.0°



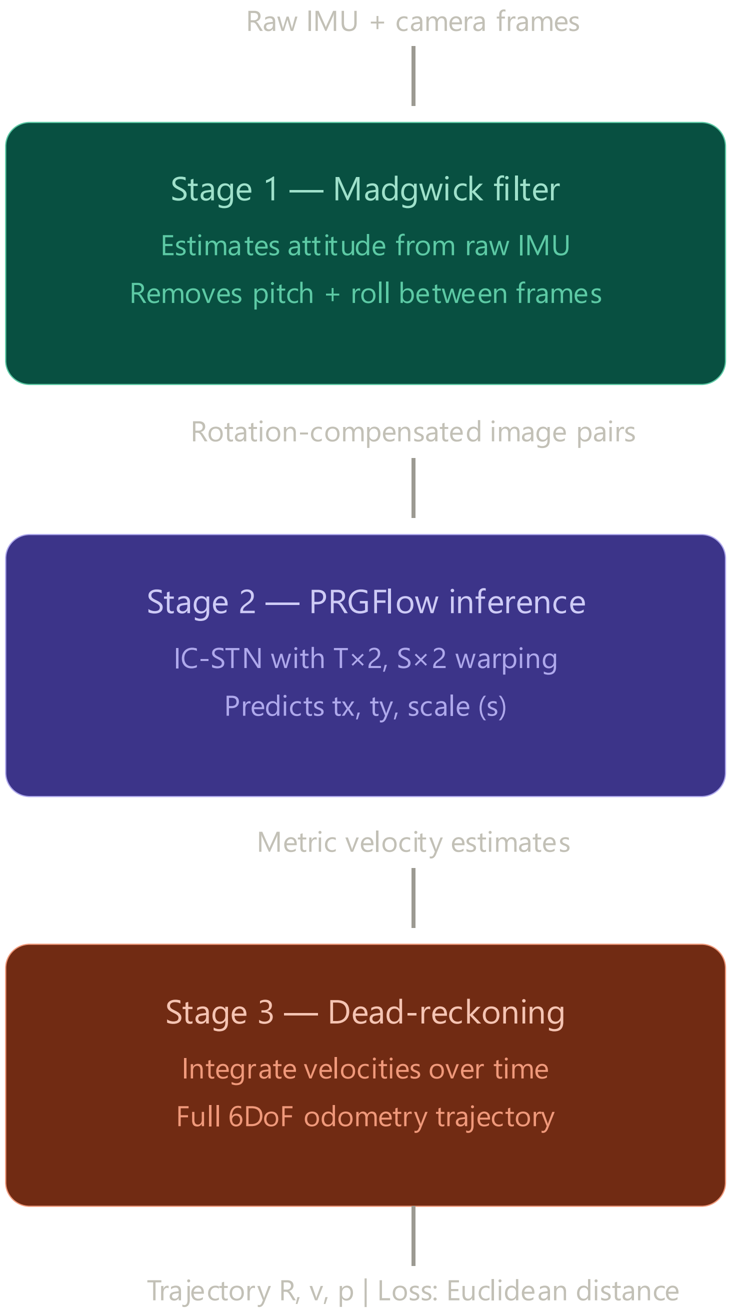
ATE: 12.73 m
RTE: 9.45 m
rot err: 103.6°

More complex training data hurt performance

AirIMU (Qiu et al., 2023) · AirIO (Qiu et al., RA-L 2025) · Forster et al., RSS 2015

VIO Architecture

Baseline implementation: PRGFlow
Stage 1: Use Madgwick filter to remove pitch and roll between consecutive frames. Only translation and scale to be estimated.
Stage 2: Compute inference on image pairs to predict 2D translation and scale. Model: SqueezeNet
Stage 3: Integrate estimated velocities over time to get the final trajectory
Loss: Euclidean distance

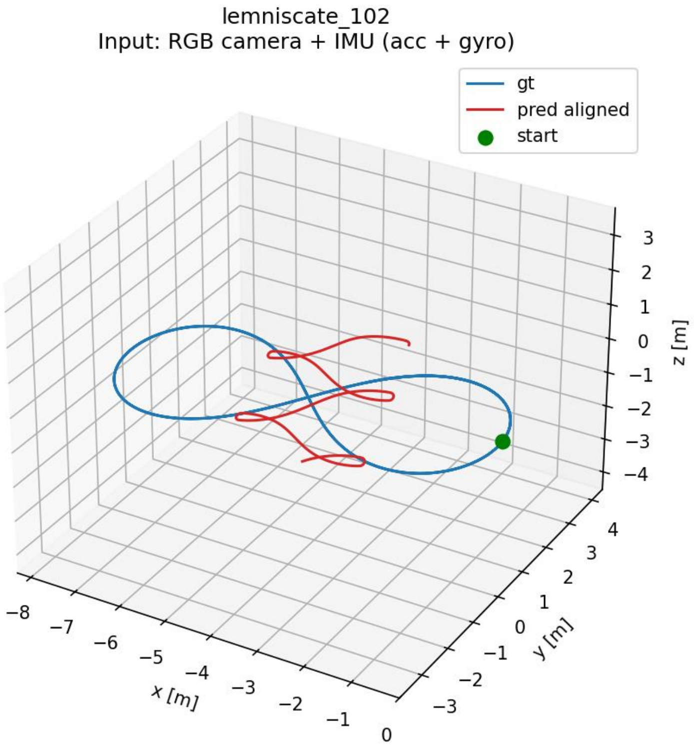


Sanket, N.J., et. al.(2021), PRGFlow: Unified SWAP-aware deep global optical flow for aerial robot navigation

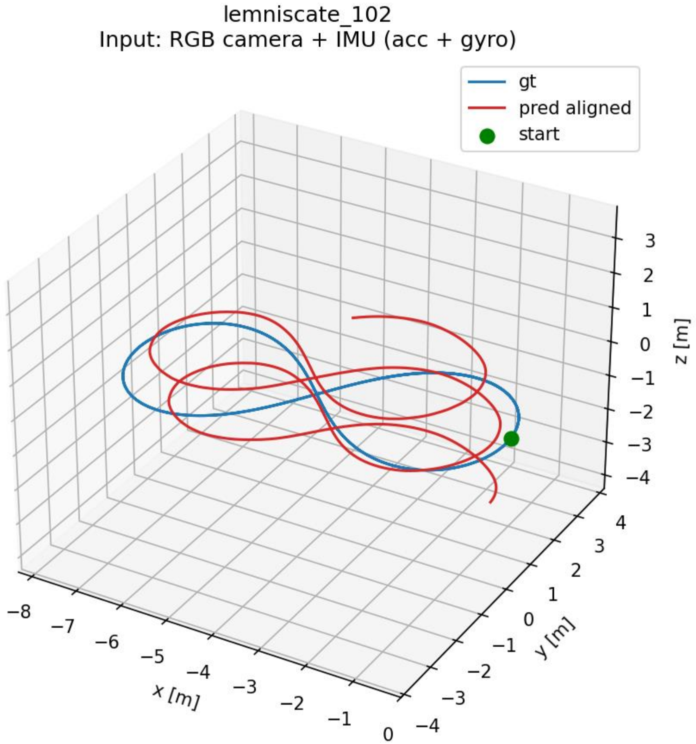
Experiments & Results

Hyperparameter modifications:
- Epochs: 100 -> 2000 due to less data
- Dropout: 0.7 -> 0.4

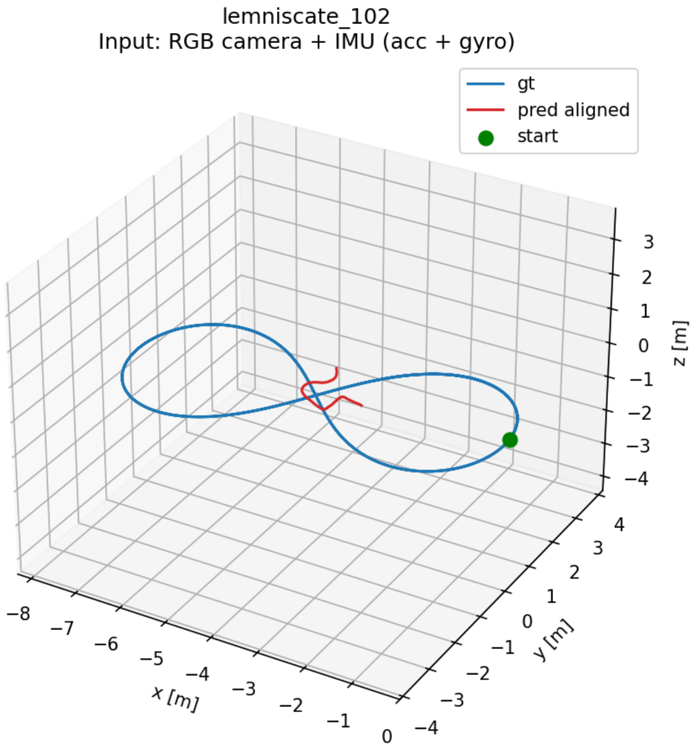
Using Geodesic loss for rotational component – Model predicts yaw as well



Initial implementation
Mean ATE = 2.1397



Hyperparameter changes
Mean ATE = 1.1759



Geodesic Loss + Yaw Prediction
Mean ATE = 2.6586

Sanket, N.J., et. al.(2021), PRGFlow: Unified SWAP-aware deep global optical flow for aerial robot navigation

Future Work

- Increasing data variation using more images with shorter trajectories.
- **Curriculum Learning:** Train on easy trajectories first, progressively introduce complexity
- **Architecture: Gated Fusion** Learned gate decides how much to trust each modality · dynamically weights vision vs IMU based on input confidence.
- **LSTM Hidden State Threading:** Hidden state now carries across chunks. Eliminates discontinuities at boundaries.
- **Hypothesis:** complementary blind spots should cancel out Camera drifts on heading. IMU drifts on position. Cross-attention should let each correct the other.